

WEEK 4

✓ SOFTWARE PROJECT MANAGEMENT – MODULE: PROJECT ESTIMATION TECHNIQUES

PART 1 – MCQ/MSQ (Assignment Related Topics)

(Questions 1–50)

1. Which of the following are *not overhead* in a software project?

- a) Staff salary
- b) Cost of building
- c) Cost of lighting
- d) Cost of networking
- e) Cost of purchasing a computer

Answer: a, e

2. In a software project, the fundamental measure of work is —

- a) Cost
- b) Size
- c) Effort
- d) Duration
- e) Time

Answer: b

3. The estimation method that compares projects to similar ones is —

- a) Parametric Models
- b) Expert Judgment
- c) Analogy-Based Method
- d) Price to Win
- e) Delphi Method

Answer: c

4. Which is *NOT true* regarding the algorithmic estimation model?

- a) It can generate repeatable estimations
- b) It is easy to modify and customize
- c) It takes less time
- d) It supports sensitivity analysis
- e) It can deal with exceptional teamwork

Answer: c, e

5. If productivity = 400 SLOC/person-month and size = 60,000 LOC, the effort is —

- a) 60 person-month
- b) 100 person-month
- c) 120 person-month
- d) 150 person-month
- e) 180 person-month

Answer: d

6. Albrecht's Function Point measure is computed from —

- a) Software Requirements Specification
- b) Design Document
- c) Source Code
- d) Configuration Document
- e) Test Report

Answer: a

7. If DI for each of 14 GSCs is 4, the VAF is —

- a) 1.07
- b) 1.21
- c) 2.07
- d) 2.21
- e) 1.31

Answer: b

8. Which is *NOT true* for Function Point Analysis?

- a) Considers algorithmic complexity
- b) Hard to fudge
- c) Language independent
- d) Understandable by clients
- e) Does not require training

Answer: a, e

9. Which are *not considered* while estimating object points?

- a) Number of screens
- b) Number of reports
- c) Number of operators
- d) Number of modules
- e) Number of operands

Answer: c, e

10. With average weights, $UFP = Inputs4 + Outputs5 + Inquiries4 + Files10 + Interfaces*7$.

If Inputs=30, Outputs=20, Inquiries=10, Files=5, Interfaces=4, UFP = ?

- a) 69
- b) 169
- c) 269
- d) 338
- e) 438

Answer: d

11. The *SPMP document* is prepared after —

- a) Requirement analysis
- b) Planning is complete
- c) Coding is done
- d) Testing is done
- e) Deployment

Answer: b

12. Overheads in a project are —

- a) Staff salaries
- b) Building, lighting, communication costs
- c) Hardware purchase
- d) Software license fees
- e) All of the above

Answer: b

13. The main factor in effort cost is —

- a) Hardware
- b) Software
- c) Engineer salaries
- d) Electricity
- e) Networking

Answer: c

14. Sliding Window Planning helps —

- a) Avoid early commitments
- b) Increase risk
- c) Reduce accuracy
- d) Delay documentation
- e) None of these

Answer: a

15. The SPMP document includes —

- a) Introduction
- b) Project Estimates
- c) Schedule
- d) Risk Management Plan
- e) All of the above

Answer: e

16. Size estimation comes before —

- a) Effort estimation
- b) Cost estimation
- c) Duration estimation
- d) Staffing estimation
- e) All of these

Answer: e

17. The cost of a project depends on —

- a) Only size
- b) Cost drivers
- c) Language used
- d) GUI complexity
- e) None

Answer: b

18. The *primary cost component* in most projects is —

- a) Hardware

- b) Effort
- c) Training
- d) Travel
- e) Network

Answer: b

19. Effort is measured in —

- a) Hours
- b) Person-weeks
- c) Person-months
- d) Person-years
- e) None

Answer: c

20. Cost and price differ because of —

- a) Market opportunity
- b) Contractual terms
- c) Requirement volatility
- d) Financial health
- e) All of the above

Answer: e

21. Parkinson's Law says —

- a) Work expands to fill the time available
- b) Work reduces with more time
- c) Work depends only on staff
- d) None

Answer: a

22. Underestimation may lead to —

- a) Higher motivation
- b) Substandard work
- c) Early delivery
- d) None

Answer: b

23. The main basis for estimation improvement is —

- a) Guesswork
- b) Historical project data
- c) Random sampling
- d) None

Answer: b

24. Estimation error can be reduced by —

- a) Refining estimates
- b) Ignoring risks
- c) Fixing plans permanently

d) Avoiding documentation

Answer: a

25. The two main estimation directions are —

a) Top-down and Bottom-up

b) Left-right

c) Inside-out

d) Past-future

Answer: a

26. Top-down estimation is useful when —

a) No detailed info available

b) Data is complete

c) Coding is finished

d) None

Answer: a

27. Bottom-up estimation is —

a) Time-consuming

b) Faster

c) Less accurate

d) Irrelevant

Answer: a

28. Algorithmic models use —

a) Size and effort relations

b) Random guessing

c) Expert bias

d) None

Answer: a

29. The most commonly used size metrics are —

a) Function Point and SLOC

b) Cost and Time

c) Duration and Effort

d) Modules and Files

Answer: a

30. LOC varies with —

a) Coding style

b) Hardware

c) Compiler

d) None

Answer: a

31. The Delphi method involves —

a) Anonymous expert rounds

b) One-time group meeting

- c) Statistical analysis
- d) Customer voting

Answer: a

32. The weighted average estimate formula is —

- a) $(O + 4M + P)/6$
- b) $(O + M + P)/3$
- c) $(O + 2M + P)/4$
- d) $(O + 5M + P)/7$

Answer: a

33. Expert judgment methods are good when —

- a) Experts have experience
- b) No experts available
- c) No past data
- d) None

Answer: a

34. “Price to Win” method —

- a) Sets cost as per customer budget
- b) Based on actual work
- c) Scientific model
- d) None

Answer: a

35. Which one is *NOT* an estimation technique?

- a) Top-down
- b) Delphi
- c) Random guess
- d) Bottom-up

Answer: c

36. Effort (E) = Size / Productivity — this is called —

- a) Simplistic model
- b) Delphi model
- c) COCOMO model
- d) Function point

Answer: a

37. Size is a measure of —

- a) Work to be done
- b) Staff strength
- c) Budget
- d) Schedule

Answer: a

38. The function point measure is based on —

- a) User’s external view
- b) Programming language
- c) Tools used

d) Hardware

Answer: a

39. Function point analysis was introduced by —

a) Albrecht

b) Boehm

c) Symons

d) Jones

Answer: a

40. The five function point parameters are —

a) EI, EO, EQ, ILF, EIF

b) Input, Output, Query, File, Code

c) Data, Module, Report, Query, Screen

d) None

Answer: a

41. The value adjustment factor formula is —

a) $(TDI \times 0.01) + 0.65$

b) $(TDI \times 0.02) + 0.5$

c) $(TDI \times 0.05) + 0.6$

d) $(TDI \times 0.1) + 0.75$

Answer: a

42. If all 14 DI = 3, VAF = ?

a) 1.07

b) 1.14

c) 1.21

d) 1.31

Answer: a

43. Function points can be estimated from —

a) Requirements

b) Design

c) Code

d) Testing

Answer: a

44. Which of the following is *NOT* a function point type?

a) External Input

b) Internal Logical File

c) External Interface File

d) Algorithmic Block

Answer: d

45. Function points are —

a) Language independent

b) Tool independent

- c) Easy to fudge
- d) Both a and b

Answer: d

46. Value Adjustment Factor (VAF) adjusts —

- a) Unadjusted Function Points
- b) Effort directly
- c) Duration
- d) Staff count

Answer: a

47. Function points were developed at —

- a) IBM
- b) Bell Labs
- c) Google
- d) NASA

Answer: a

48. Which one is a *disadvantage* of SLOC?

- a) Varies with style
- b) Ignores complexity
- c) Hard to estimate early
- d) All of the above

Answer: d

49. Function points overcome SLOC issues because they are —

- a) Language independent
- b) Early measurable
- c) User oriented
- d) All of the above

Answer: d

50. Function points mainly measure —

- a) Functionality delivered
- b) LOC written
- c) Time spent
- d) Cost incurred

Answer: a

✓ **SOFTWARE PROJECT MANAGEMENT – MODULE: PROJECT ESTIMATION TECHNIQUES**

PART 2 – MCQ/MSQ (Remaining Module Topics)

(Questions 51–75)

51. Mark II Function Points were developed by —

- a) Albrecht
- b) Boehm
- c) Symons
- d) Jones
- e) COSMIC Group

Answer: c

52. Mark II FP formula is —

- a) $N_i \times 0.58 + N_e \times 1.66 + N_o \times 0.26$
- b) $N_i \times 1.2 + N_e \times 0.5 + N_o \times 0.25$
- c) $N_i \times 0.75 + N_e \times 1.5 + N_o \times 0.5$
- d) $N_i \times 0.60 + N_e \times 1.00 + N_o \times 0.40$

Answer: a

53. COSMIC Function Points are suitable for —

- a) Embedded systems
- b) Business systems
- c) Web applications only
- d) None

Answer: a

54. COSMIC Function Points count data movements as —

- a) Entry, Exit, Read, Write
- b) Input, Output, Process, File
- c) Create, Update, Delete, View
- d) None

Answer: a

55. Each data movement in COSMIC counts as —

- a) 1 COSMIC functional size unit
- b) 10 units
- c) 5 units
- d) 0.5 units

Answer: a

56. Function points for embedded systems are based on —

- a) Layers of software
- b) Database size
- c) Lines of code
- d) GUI

Answer: a

57. The Feature Point metric adds which extra parameter to FP?

- a) Algorithm complexity
- b) File size
- c) Execution speed
- d) Data storage

Answer: a

58. Feature Point metric was introduced to overcome —

- a) Complexity independence of FP
- b) Training cost
- c) Tool dependency
- d) Size variation

Answer: a

59. FP to SLOC conversion varies depending on —

- a) Programming language
- b) Project manager
- c) Customer location
- d) None

Answer: a

60. In the FP/SLOC conversion, 1 FP = 59 SLOC for —

- a) Java
- b) C
- c) C++
- d) Perl

Answer: a

61. Function Points are best used for —

- a) Estimating software size
- b) Designing architecture
- c) Scheduling tasks
- d) Testing automation

Answer: a

62. Object Points are based on —

- a) Screens, reports, and modules
- b) Files, inputs, and outputs
- c) Algorithms and loops
- d) Tasks and subtasks

Answer: a

63. Object Points do *not* include —

- a) Screens
- b) Reports
- c) Modules
- d) Operators

Answer: d

64. Function Point metric focuses on —

- a) Functionality from user's perspective
- b) Internal logic
- c) Code structure
- d) Algorithms only

Answer: a

65. In Function Point counting, ILF stands for —

- a) Internal Logical File
- b) Integrated Log Function
- c) Internal Line Format
- d) Input Logical Form

Answer: a

66. EIF in FP analysis refers to —

- a) External Interface File
- b) External Information Form
- c) External Input File
- d) Extended Internal Function

Answer: a

67. UFP means —

- a) Unadjusted Function Points
- b) Uniform Function Process
- c) Unique Function Parameter
- d) Unspecified File Program

Answer: a

68. The full form of VAF is —

- a) Value Adjustment Factor
- b) Variable Adjustment Formula
- c) Verified Analysis Factor
- d) Version Adjustment Function

Answer: a

69. The typical range of VAF is —

- a) 0.65 to 1.35
- b) 1.0 to 2.0
- c) 0.50 to 1.50
- d) 1.5 to 3.0

Answer: a

70. Function Points can be applied during —

- a) Requirement analysis phase
- b) Only testing
- c) Only coding
- d) Only maintenance

Answer: a

71. COSMIC FPs were standardized by —

- a) ISO
- b) IEEE
- c) IBM
- d) Oracle

Answer: a

72. In estimation, “calibration” means —
- a) Adjusting model for specific environment
 - b) Increasing project cost
 - c) Ignoring risk
 - d) Random sampling

Answer: a

73. In a payroll system, the input of employee details is —
- a) External Input
 - b) External Output
 - c) External Inquiry
 - d) Internal Logical File

Answer: a

74. In a software system, the report generated for users is —
- a) External Output
 - b) Internal File
 - c) Interface
 - d) Inquiry

Answer: a

75. The total Function Points are calculated as —
- a) $FP = UFP \times VAF$
 - b) $FP = UFP + VAF$
 - c) $FP = UFP / VAF$
 - d) $FP = VAF - UFP$

Answer: a

✓ **END OF PART 2 (Q51–75)**